

Deploy a Web Application to Kubernetes using Minikube.

STEP 1 - Start Minikube

minikube start

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~$ minikube start
🐸 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" .../
```

STEP 2 - Check Minikube Status

minikube status

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

STEP 3 - If Minikube is not running, restart it

minikube delete

minikube start --driver=docker

STEP 4 - Create the Deployment file (nginx-deployment.yaml)

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ cat
nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

STEP 5 - Apply the deployment

kubectl apply -f nginx-deployment.yaml\

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

STEP 6 - Check Deployment & Pods

kubectl get deployments

kubectl get pods

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2            2           15s
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-g7rsn  1/1     Running   0          20s
nginx-deployment-6f9664446b-vrcvl  1/1     Running   0          20s
```

You should see 2 pods running.

STEP 7 -Expose the Deployment as a Service

kubectl expose deployment nginx-deployment \

--type=NodePort \

--port=80

This creates a service that exposes Nginx to your system.

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80
service/nginx-deployment exposed
```

STEP 8 - Check Services

kubectl get services

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ kubectl get svc
NAME                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP    10.96.0.1    <none>        443/TCP          6d23h
nginx-deployment    NodePort     10.105.3.180 <none>        80:31602/TCP     14s
```

A service named nginx-deployment should appear .

STEP 9 - Access the Webpage

Minikube provides a shortcut to open the app:

minikube service nginx-deployment

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31602

```
🐚 Opening service default/nginx-deployment in default browser...
```

This will open the browser showing the Nginx Welcome Page.

⚠ Not secure 192.168.49.2:31602

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

STEP 10 — Clean Up

(Optional)

Delete all pods

```
kubectl delete pods --all
```

Delete all services

```
kubectl delete svc --all
```

Delete all deployments

```
kubectl delete deployments --all
```

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ kubectl delete svc --all
service "kubernetes" deleted
service "nginx-deployment" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-g7rsn" deleted
pod "nginx-deployment-6f9664446b-vrcvl" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program7$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted
```